

# CuocoGPT



A Bilingual Recipe Assistant Combining a Knowledge Graph,  
Lightweight NLP, and LLMs

Presented by Miguel {Castela, Martins}



# Objective & Challenges

## Objective

- Develop a Bilingual Assistant that understands intent.
- Combine the precision of Knowledge Graphs (KG) with the fluency of LLMs.

## Challenges:

- Recipe retrieval is often limited to keyword matching.
- Users have complex, multi-criteria constraints (dietary, time, ingredients).
- Multilingual support (PT/EN) is challenging for structured data.

# System Pipeline

- **Knowledge Graph:** Construction of the knowledge graph using RDFS
- **NLP Layer:** Intent classification (BERT) + Entity Extraction (spaCy/RapidFuzz).
- **Retrieval:** SPARQL for structured data + FAISS index on descriptions.
- **Generation:** LLM (Groq API) generates summarized results bilingually.

02

# Dataset and Knowledge Graph

- **Dataset:** Portion of the **Food.com** dataset, both manually and computationally filtered, with 10.000 entries. **10%** are Portuguese and the others are randomly selected.
- **Knowledge Graph:** Generated using **RDFS**, with classes such as recipe or ingredient and fields such as minutes or id.
- **Pre-loading:** Some heavy components such as the knowledge graph and spaCy model are loaded using a **Pickle** snapshot.

03





# Fuzzy Search

find\_recipe → "receita de arroz de pato" → top recipe names

retrieve\_ingredients → "o que preciso para fazer pizza" → top recipe names

get\_prep\_time → "how long to cook for lasagna" → top recipe names

list\_by\_time → "receitas em 10 minutos" → minutes (exact or less then)

list\_by\_ingredient → "recipes that include pork" → Ingredient name

list\_by\_tag → "find low-carb meals" → top tags

05



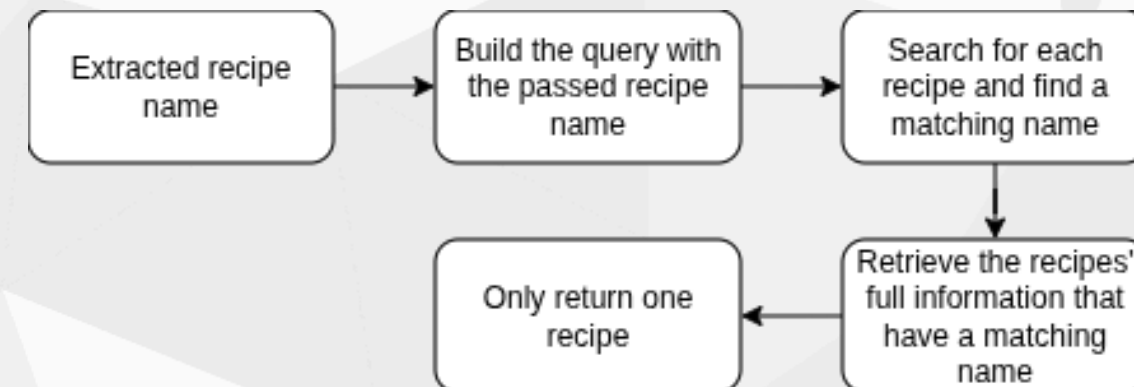
# SPARQL Functions

Each user intent is mapped to a specific SPARQL query function, filled with extracted entities, to retrieve structured data from the KG.

## retrieve\_ingredients



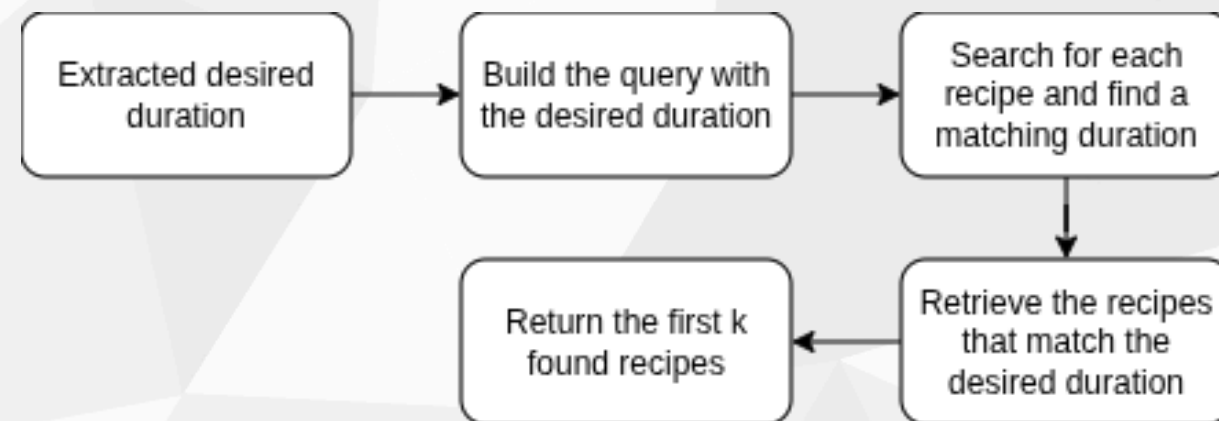
## find\_recipe



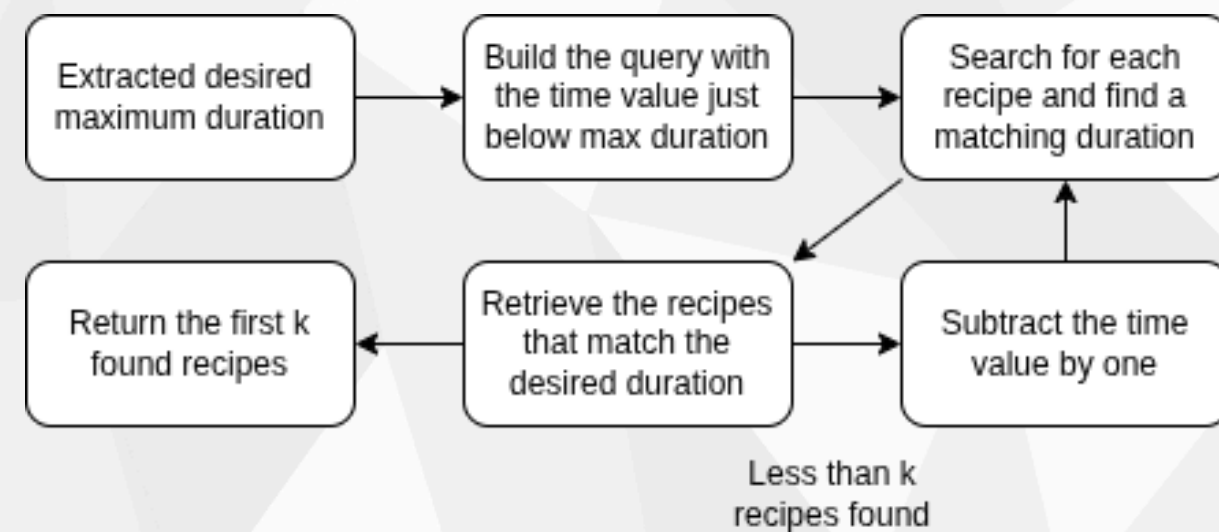
# SPARQL Functions

**list\_by\_time** (if hour or minutes value is matched):

get\_exact\_time (ex: "in 30 minutes")



get\_max\_time (ex: "under 30 minutes")

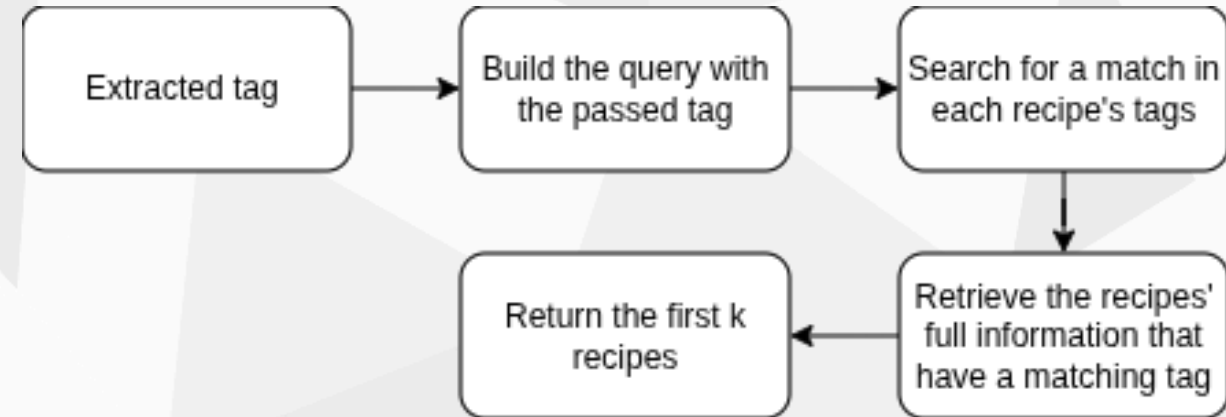


# 07

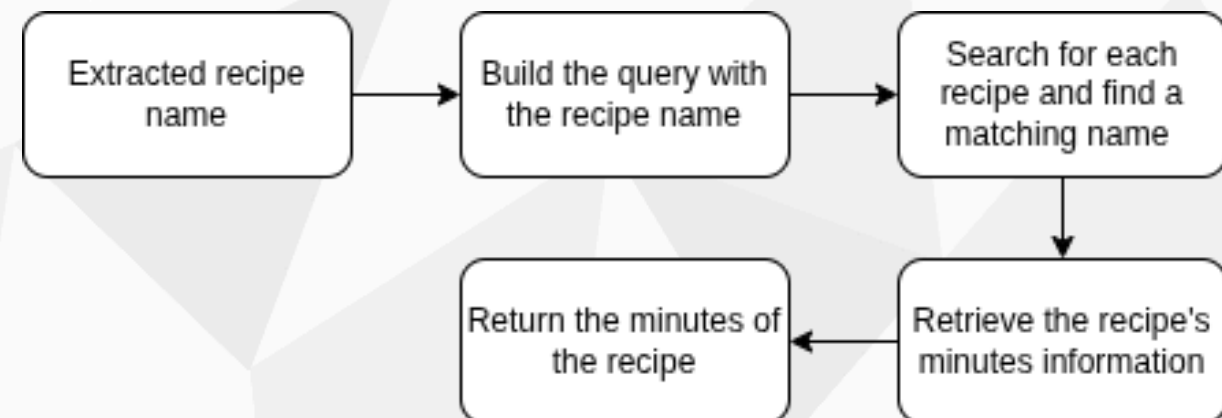


# SPARQL Functions

**list\_by\_tag** / list\_by\_time fuzzy fallback



**Get\_prep\_time**



**List\_by\_ingredient**



# Experimental Results

## Key Findings

- **Bilingual Consistency:** Improved translation normalization ensured stable and deterministic results across PT/EN queries.
- **RAG Impact:** RAG implementation in the descriptions with FAISS showed little measurable impact due to the high ambiguity of the texts, even after removing outliers.
- **Hard Queries:** For creative queries, the LLM used retrieved data to ask clarifying questions, guiding the user back to valid intents.

CuocoGPT

-

12/2025

09

# Conclusion & Future Work

## Summary:

- Successfully integrated a KG with LLMs for a robust bilingual assistant.
- Solved specific NLP challenges in entity linking, time parsing, and translation consistency

## Future Directions

- Expand KG: Include more recipes and nutritional facets.
- Personalization: Implement user profiles for dietary restrictions.

# Thank You

